ATAR · YEAR 12 CHEMISTRY · UNIT 3

Redox & Electrochemistry

Complete Study Notes

Definitions · oxidation numbers · half-equations · galvanic & electrolytic cells · standard electrode potentials · Faraday calculations · corrosion · redox titrations · exam traps · full data sheet.

1 Redox Fundamentals

A **redox reaction** involves the transfer of electrons between species. **Oxidation** and **reduction** always happen together — the electrons lost by one species are gained by another. Electrons are never created or destroyed.

OIL RIG

Oxidation **I**s **L**oss · **R**eduction **I**s **G**ain (of electrons)

Process	Electrons	Oxidation number	Oxygen	Hydrogen
Oxidation	lost	increases	gained	lost
Reduction	gained	decreases	lost	gained

The electron-transfer definition is the most general and is the one to use at ATAR level. The O/H definitions are handy shortcuts for combustion and organic reactions.

2 Oxidation Numbers (States)

The **oxidation number** is the charge an atom *would* have if every bond were fully ionic. It is the bookkeeping tool for tracking electron transfer and identifying what is oxidised or reduced.

> **RULES — APPLY IN THIS PRIORITY ORDER**
> 1. Free element = **0** (e.g. Na, O_2, Cl_2, S_8, P_4).
> 2. Monatomic ion = its **charge** (Na^+=+1, S^{2-}=−2, Al^{3+}=+3).
> 3. Group 1 = +1; Group 2 = +2; Al = +3 (in compounds).
> 4. F = −1 always. H = +1 (but −1 in metal hydrides, e.g. NaH).
> 5. O = −2 (but −1 in peroxides e.g. H_2O_2; +2 in OF_2).
> 6. Sum = **0** in a neutral compound; = **ion charge** in a polyatomic ion.

WORKED EXAMPLES — SOLVE FOR THE UNKNOWN

SO_4^{2-}: $x + 4(-2) = -2 \implies$ **$S = +6$**

MnO_4^{-}: $x + 4(-2) = -1 \implies$ **$Mn = +7$**

$Cr_2O_7^{2-}$: $2x + 7(-2) = -2 \implies$ **$Cr = +6$**

NH_4^{+}: $x + 4(+1) = +1 \implies$ **$N = -3$**

USING OXIDATION NUMBERS TO IDENTIFY REDOX

Compare each element before and after: an **increase** = oxidised; a **decrease** = reduced; **no change** = not a redox reaction (e.g. most acid–base and precipitation reactions).

Disproportionation

The *same* element is simultaneously oxidised *and* reduced:

$Cl_2 + H_2O \rightarrow HOCl + HCl$ (Cl: $0 \rightarrow +1$ and $0 \rightarrow -1$)

3 Oxidants, Reductants & Conjugate Pairs

DEFINITIONS

- **Oxidising agent (oxidant):** accepts electrons → is itself **reduced**.
- **Reducing agent (reductant):** donates electrons → is itself **oxidised**.

COMMON CONFUSION

The oxidising agent is the species that gets **reduced**; the reducing agent is the species that gets **oxidised**. Read it twice.

Each pair is linked by electron transfer — a **conjugate redox pair**:

$$\textbf{oxidised form} + n\,e^- \rightleftharpoons \textbf{reduced form}$$

e.g. Cu^{2+}/Cu and Zn^{2+}/Zn. The stronger an oxidant, the weaker its conjugate reductant.

Common species at ATAR

Strong oxidising agents	Strong reducing agents
F_2, MnO_4^-/H^+, $Cr_2O_7^{2-}/H^+$, Cl_2, H_2O_2, conc. HNO_3	Reactive metals (Li, K, Na, Mg, Al, Zn), H_2, C, I^-, SO_2

4 Writing & Balancing Half-Equations

A **half-equation** shows one process (oxidation or reduction) with its electrons. Electrons appear on the **left** of a reduction and on the **right** of an oxidation.

BALANCING IN ACIDIC SOLUTION — 5 STEPS

1. Balance all atoms **except O and H**.
2. Balance **O** with **H_2O**.
3. Balance **H** with **H^+**.
4. Balance **charge** with **e^-**.
5. Check atoms *and* charge.

EXAMPLE — PERMANGANATE REDUCED IN ACID

$MnO_4^- \rightarrow Mn^{2+} + 4H_2O \;\rightarrow\;$ add $8H^+$ $\;\rightarrow\;$ add $5e^-$ (left, +7→+2)

$$\mathbf{MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O}$$

BASIC (ALKALINE) SOLUTION

Balance as if acidic, then add **OH^- to both sides** equal to the H^+ present. Combine $H^+ + OH^- \rightarrow H_2O$ and cancel any water appearing on both sides.

5 Combining Half-Equations

METHOD

1. Write the oxidation and reduction half-equations.
2. Multiply each so the **electrons are equal**.
3. Add them; **cancel electrons** and any identical species on both sides.

EXAMPLE — MNO_4^- OXIDISING FE^{2+} IN ACID

Reduction (×1): $MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$

Oxidation (×5): $5Fe^{2+} \rightarrow 5Fe^{3+} + 5e^-$

Overall: $\mathbf{MnO_4^- + 8H^+ + 5Fe^{2+} \rightarrow Mn^{2+} + 5Fe^{3+} + 4H_2O}$

Charge check: left (−1+8+10)=+17; right (+2+15)=+17 ✓

6 Galvanic (Voltaic) Cells

A galvanic cell converts **chemical → electrical** energy using a **spontaneous** redox reaction ($E°_{cell} > 0$). Separating the two half-reactions forces electrons through an external wire — this is the current.

Component	Role
Anode	**Oxidation** occurs here. Galvanic → **negative** terminal.
Cathode	**Reduction** occurs here. Galvanic → **positive** terminal.
Salt bridge	Inert electrolyte (e.g. KNO_3); completes the circuit and keeps each half-cell **electrically neutral** by allowing ion flow. It does *not* carry electrons.
External wire	Carries electrons anode → cathode; holds the voltmeter/load.

MEMORY AIDS

- **AN OX** = ANode is OXidation. **RED CAT** = REDuction is CAThode.
- Electrons always flow **anode → cathode** in the wire.
- In the salt bridge: **anions → anode**, **cations → cathode**.
- Galvanic polarity: anode **−**, cathode **+** (reversed in electrolysis).

DANIELL CELL (ZN | CU)

Anode: $Zn(s) \rightarrow Zn^{2+}(aq) + 2e^-$ ($E° = -0.76$ V)

Cathode: $Cu^{2+}(aq) + 2e^- \rightarrow Cu(s)$ ($E° = +0.34$ V)

Overall: $Zn(s) + Cu^{2+}(aq) \rightarrow Zn^{2+}(aq) + Cu(s)$ $E°_{cell} = +1.10$ V

Zn electrode loses mass; Cu electrode gains mass; $[Cu^{2+}]$ falls, $[Zn^{2+}]$ rises.

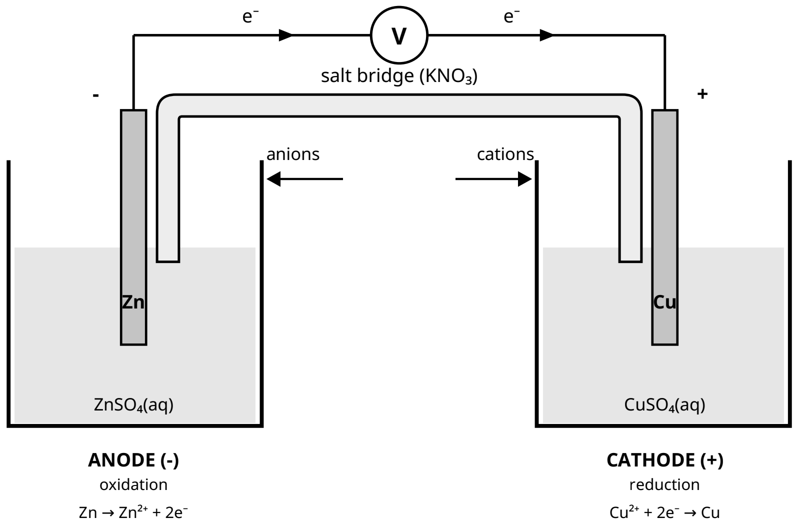

Figure 1. Galvanic (Daniell) cell. Electrons flow through the wire from the Zn anode (−) to the Cu cathode (+); the salt bridge carries ions to keep each half-cell neutral (anions → anode, cations → cathode).

Cell diagram notation

Anode on the left. Single line **|** = phase boundary; double line **‖** = salt bridge.

$$Zn(s) \mid Zn^{2+}(aq) \parallel Cu^{2+}(aq) \mid Cu(s)$$

If a half-cell has no solid conductor (e.g. Fe^{3+}/Fe^{2+}) use an **inert electrode** (Pt or graphite): Pt | Fe^{2+}, Fe^{3+}.

7 Standard Electrode Potentials

Each half-cell's tendency to gain electrons is its **standard reduction potential (E°)**, measured in volts against the **Standard Hydrogen Electrode (SHE)**, defined as exactly **0.00 V**.

STANDARD CONDITIONS (THE "°")

Solutions **1.0 mol L^{-1}** · gases **100 kPa** · temperature **25 °C (298 K)** · measured vs the SHE.

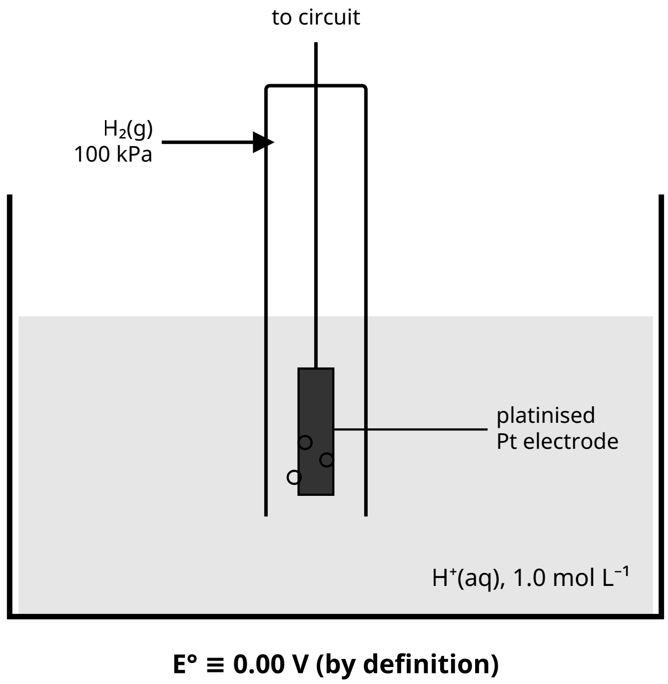

Figure 2. The Standard Hydrogen Electrode (SHE): H_2 gas at 100 kPa bubbled over an inert platinised Pt electrode in 1.0 mol L^{-1} H^+, at 25 °C. Half-cell: $2H^+ + 2e^- \rightleftharpoons H_2$; assigned E° = 0.00 V so all other potentials can be measured against it.

READING THE TABLE (ALL VALUES ARE REDUCTIONS)

- **More positive E°** → stronger **oxidising agent** (top-left of table; e.g. F_2).
- **More negative E°** → its reduced form is a stronger **reducing agent** (bottom-right; e.g. Li).

PREDICTION RULE ("ANTICLOCKWISE")

Place the two half-equations with the more positive E° on top. The **top reaction goes forward (reduction)** and the **bottom reverses (oxidation)**: the oxidant (top-left) reacts with the reductant (bottom-right).

8 Cell EMF & Predicting Reactions

STANDARD CELL POTENTIAL

$$E^\circ_{cell} = E^\circ_{cathode} - E^\circ_{anode}$$

Use both values straight from the table as reductions — the minus sign handles the reversal; do **not** flip the sign yourself.

SIGN OF E°_{CELL}

- **$E^\circ_{cell} > 0$** → spontaneous (works as a galvanic cell); $\Delta G < 0$.
- **$E^\circ_{cell} < 0$** → non-spontaneous (needs electrolysis).

Free-energy link: $\Delta G^\circ = -nFE^\circ_{cell}$ (n = mol e^-, F = 96 500 C mol^{-1}).

EXAMPLE — DOES COPPER METAL REACT WITH AG^+?

$Ag^+ + e^- \rightarrow Ag$ (+0.80 V) is reduced (cathode); $Cu \rightarrow Cu^{2+} + 2e^-$ is oxidised (anode).

$E^\circ_{cell} = (+0.80) - (+0.34) = \mathbf{+0.46\ V}$ → spontaneous

$Cu(s) + 2Ag^+(aq) \rightarrow Cu^{2+}(aq) + 2Ag(s)$

E° DOES NOT SCALE WITH AMOUNT

E° is **intensive**. When you multiply a half-equation to balance electrons, do **not** multiply its E° value.

Limitations of E° predictions

- Only valid at **standard conditions**; concentration, temperature and pressure shift the real potential.
- Predicts **feasibility, not rate** — a spontaneous reaction may be extremely slow.
- **Overpotential** can change the observed product (see §10).

9 Electrolytic Cells

Electrolysis uses an external power supply to drive a **non-spontaneous** redox reaction ($E°_{cell} < 0$): electrical → chemical energy — the reverse of a galvanic cell.

Feature	Galvanic	Electrolytic
Energy	chemical → electrical	electrical → chemical
Spontaneity	spontaneous (E°>0)	non-spontaneous (E°<0)
Power supply	none (is the source)	required
Anode sign	negative (−)	**positive (+)**
Cathode sign	positive (+)	**negative (−)**

WHAT STAYS THE SAME

Oxidation is **always** at the anode and reduction **always** at the cathode; electrons **always** travel anode → cathode. Only the **polarity (+/−) flips** between cell types.

MOLTEN NACL (DOWNS PROCESS)

$$\text{Cathode } (-)\text{: } Na^+ + e^- \rightarrow Na(l) \quad \text{Anode } (+)\text{: } 2Cl^- \rightarrow Cl_2(g) + 2e^-$$

No water present, so only the salt's ions react.

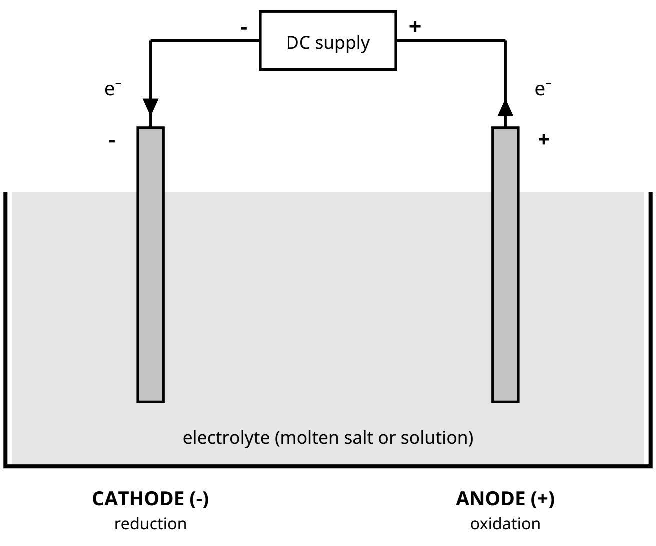

Figure 3. Electrolytic cell. An external DC supply forces electrons to the cathode, driving a non-spontaneous reaction. Note the polarity is reversed vs a galvanic cell (anode +, cathode −), but oxidation is still at the anode and reduction still at the cathode.

10 Predicting Electrolysis Products (Aqueous)

In water, **H_2O itself can be oxidised or reduced**, competing with the dissolved ions:

Cathode: $2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$ ($E° = -0.83$ V)

Anode: $2H_2O \rightarrow O_2(g) + 4H^+ + 4e^-$ (reduction E° of $O_2/H_2O = +1.23$ V)

DECISION PROCEDURE AT EACH ELECTRODE

1. List every species that could react there (ions + water).
2. **Cathode:** the species with the **highest E°** is reduced.
3. **Anode:** the species with the **lowest (most negative) E°** is oxidised.
4. Then apply the practical factors below.

Factor	Effect on product
E° value	The primary predictor.
Concentration	A high ion concentration can cause it to discharge in preference to water (e.g. concentrated $Cl^- \rightarrow Cl_2$).
Overpotential	Extra voltage needed to release **gases** (especially O_2); this is why Cl_2 often forms instead of O_2.
Electrode type	Inert (Pt, C) is unchanged; an **active** electrode (e.g. Cu) may itself be oxidised at the anode.

EXAMPLE — AQUEOUS $CUSO_4$, INERT ELECTRODES

Cathode: Cu^{2+} (+0.34) vs H_2O (−0.83) → **Cu(s) deposited**

Anode: SO_4^{2-} very hard to oxidise → water oxidised → **O_2(g)** + H^+

Copper plates out; O_2 bubbles at the anode; solution becomes acidic.

11 Faraday's Laws — Electrolysis Calculations

The amount of substance produced at an electrode is proportional to the charge passed.

FORMULAE & 5-STEP METHOD

Formula	Terms
$Q = I \times t$	Q charge (C); I current (A); t time (**seconds**)
$n(e^-) = Q \div F$	$F = 96\,500$ C mol^{-1}
half-equation ratio	relate mol e^- → mol product
$m = n \times M$ / $V = n \times V_m$	convert to mass or gas volume

Steps: (1) Q = It → (2) $n(e^-) = Q/96\,500$ → (3) read electron:product ratio → (4) find mol product → (5) convert to mass/volume.

EXAMPLE A — SILVER ELECTROPLATING

2.00 A for 30.0 min through $AgNO_3$. Mass of Ag deposited?

$Q = 2.00 \times (30.0\times60) = 3600$ C

$n(e^-) = 3600 \div 96\,500 = 0.03731$ mol

$Ag^+ + e^- \rightarrow Ag$ (1:1) $\rightarrow n(Ag) = 0.03731$ mol

$m = 0.03731 \times 107.9 =$ **4.03 g Ag**

EXAMPLE B — GAS VOLUME

Same 3600 C in electrolysis of water; V of H_2 at 25 °C, 100 kPa ($V_m = 24.79$ L mol^{-1})?

$2H_2O + 2e^- \rightarrow H_2 + 2OH^-$ (2:1) $\rightarrow n(H_2) = 0.03731 \div 2 = 0.01865$ mol

$V = 0.01865 \times 24.79 =$ **0.463 L (463 mL)**

CALCULATION TRAPS

Convert time to **seconds**; watch the **electron ratio** (Cu^{2+} needs $2e^-$, Al^{3+} needs $3e^-$); cells **in series** pass the **same charge**.

12 Industrial Applications

Electroplating

Object = **cathode**; pure plating metal = **anode**; electrolyte contains that metal's ions.

Cathode: $M^{n+} + n\,e^- \rightarrow M(s)$ Anode: $M(s) \rightarrow M^{n+} + n\,e^-$

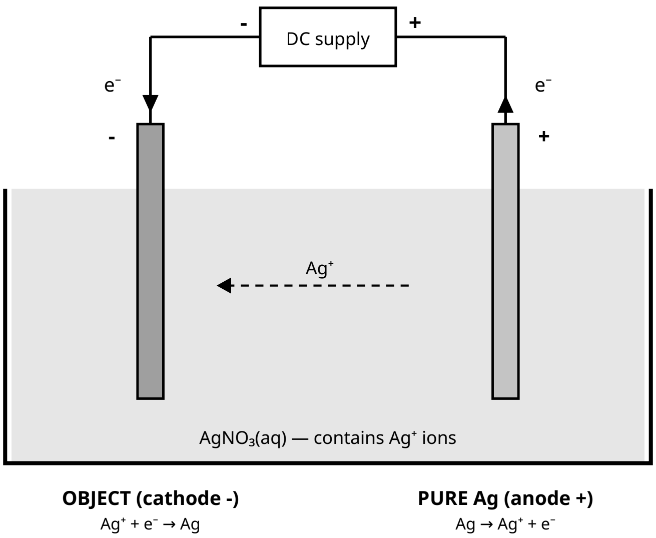

Figure 4. Electroplating with silver. The object is the cathode (Ag^+ is reduced onto it); the pure silver anode dissolves to replenish Ag^+, so the solution concentration stays roughly constant.

Electrorefining of copper

Impure Cu = anode (dissolves); pure Cu deposits on the cathode. Ag/Au fall as "anode sludge"; more reactive metals stay in solution.

Extraction of reactive metals (aluminium — Hall–Héroult)

Very reactive metals (Al, Na, Mg, K) are obtained by electrolysing the **molten** compound. Al_2O_3 is dissolved in molten cryolite to lower the melting point.

$$\text{Cathode: } Al^{3+} + 3e^- \rightarrow Al(l) \quad \text{Anode: } 2O^{2-} \rightarrow O_2 + 4e^- \text{ (carbon anodes burn to } CO_2\text{)}$$

13 Corrosion of Iron & Its Prevention

Corrosion is unwanted oxidation of a metal. Rusting of iron is electrochemical and needs **both water and oxygen**; it is faster with dissolved salts (electrolytes) or acids.

THE ELECTROCHEMISTRY OF RUSTING

$$\text{Anodic: } Fe(s) \rightarrow Fe^{2+}(aq) + 2e^-$$

$$\text{Cathodic: } O_2(g) + 2H_2O(l) + 4e^- \rightarrow 4OH^-(aq)$$

Fe^{2+} is further oxidised by O_2 to hydrated iron(III) oxide, $Fe_2O_3{\cdot}xH_2O$ = rust.

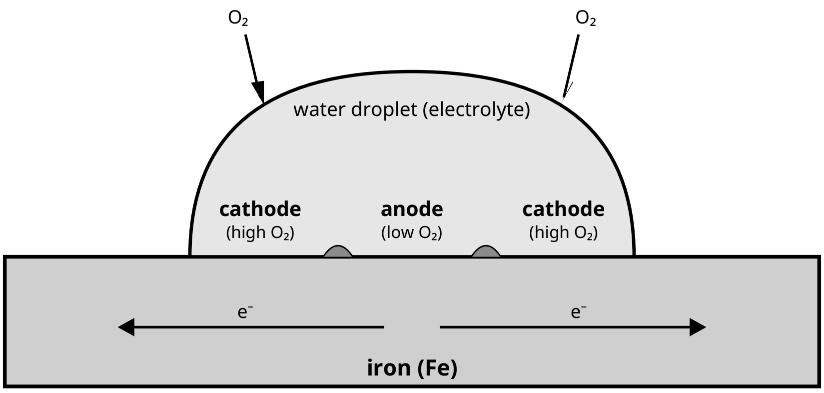

Figure 5. Rusting of iron under a water droplet. Centre (low O_2) = anode: $Fe \rightarrow Fe^{2+} + 2e^-$. Edges (high O_2) = cathode: $O_2 + 2H_2O + 4e^- \rightarrow 4OH^-$. Electrons travel through the metal; Fe^{2+} and OH^- meet to deposit rust, $Fe_2O_3{\cdot}xH_2O$.

Method	How it works
Barrier (paint, oil, plastic)	Excludes water/O_2; fails if scratched.
Galvanising (Zn coat)	Barrier *plus* sacrificial protection — even if scratched, Zn oxidises before Fe.
Sacrificial anode (Zn/Mg block)	A more reactive metal is oxidised instead of the iron (ships, pipelines); replaced periodically.
Cathodic protection	External supply forces the iron to be the **cathode**, so it cannot oxidise.
Alloying (stainless steel)	Cr/Ni form a self-repairing protective oxide layer.

WHY SACRIFICIAL PROTECTION WORKS

A metal with a **more negative E°** than iron (Zn −0.76, Mg −2.37 vs Fe −0.44) is a stronger reducing agent, so it is oxidised preferentially and protects the iron even where the coating breaks.

14 Redox Titrations

A redox titration finds an unknown concentration by reacting it completely with a standard oxidant/reductant, using the balanced equation's mole ratio.

PERMANGANATE TITRATIONS (SELF-INDICATING)

Acidified MnO_4^- (deep purple) → Mn^{2+} (colourless). The **endpoint** is the first **permanent faint pink** when one drop of excess MnO_4^- remains — no indicator needed.

WORKED EXAMPLE

25.0 mL of Fe^{2+} needs 20.0 mL of 0.0200 mol L^{-1} $KMnO_4$. Find $[Fe^{2+}]$.

$$n(MnO_4^-) = 0.0200 \times 0.0200 = 4.00\times10^{-4}\ \text{mol}$$

$$\text{Ratio } MnO_4^-:Fe^{2+} = 1:5 \rightarrow n(Fe^{2+}) = 2.00\times10^{-3}\ \text{mol}$$

$$[Fe^{2+}] = 2.00\times10^{-3} \div 0.0250 = \mathbf{0.0800\ mol\ L^{-1}}$$

Iodine/thiosulfate titrations ($I_2 + 2S_2O_3^{2-} \rightarrow 2I^- + S_4O_6^{2-}$) use a **starch** indicator; endpoint = blue-black just disappears.

15 Batteries & Fuel Cells

Type	Description & examples
Primary	Non-rechargeable; reaction not reversible (dry cell Zn/MnO_2, alkaline).
Secondary	Rechargeable — recharging is electrolysis that reverses the reaction (lead–acid, lithium-ion).
Fuel cell	Reactants supplied continuously; high efficiency, low pollution (H_2/O_2).

HYDROGEN–OXYGEN FUEL CELL (ACIDIC)

$$\text{Anode: } 2H_2 \rightarrow 4H^+ + 4e^- \quad \text{Cathode: } O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$$

$$\text{Overall: } 2H_2 + O_2 \rightarrow 2H_2O$$

16 Galvanic vs Electrolytic — Summary

	Galvanic (voltaic)	Electrolytic
Purpose	generates electricity	uses electricity to drive a reaction
Energy	chemical → electrical	electrical → chemical
ΔG / $E°_{cell}$	ΔG<0, E°>0 (spontaneous)	ΔG>0, E°<0 (non-spontaneous)
Power supply	none	required
Oxidation / Reduction	anode / cathode	anode / cathode
Anode sign	negative (−)	positive (+)
Cathode sign	positive (+)	negative (−)
Electron flow	anode → cathode	anode → cathode
Examples	batteries, fuel cells	electroplating, refining, Al/NaCl extraction, recharging

17 Exam Traps & Glossary

TOP TRAPS TO AVOID

- Oxidising agent is **reduced**; reducing agent is **oxidised**.
- Never change E° when multiplying a half-equation (E° is intensive).
- Salt bridge maintains **neutrality/completes the circuit** — it does not carry electrons.
- Only the **+/−** flips between cell types; oxidation is always at the anode.
- Convert time to **seconds** in Q = It; use the correct **electron ratio**.
- Include **water** as a competitor in aqueous electrolysis.
- Always **check charge balance** in half-equations.
- Positive $E°_{cell}$ means **feasible**, not fast.

Glossary

Term	Definition
Redox reaction	Electron-transfer reaction (simultaneous oxidation + reduction).
Oxidation / Reduction	Loss / gain of electrons (ox. number increases / decreases).
Oxidising agent	Electron acceptor; is itself reduced.
Reducing agent	Electron donor; is itself oxidised.
Oxidation number	Charge an atom would have if all bonds were ionic.
Half-equation	Shows one process with its electrons.
Conjugate redox pair	Oxidised and reduced forms linked by e^- transfer.
Anode / Cathode	Electrode of oxidation / reduction.
Salt bridge	Maintains neutrality and completes a galvanic circuit.
E° (standard potential)	Reduction potential of a half-cell vs the SHE at standard conditions.
SHE	Standard Hydrogen Electrode; reference defined as 0.00 V.
Electrolysis	Using electrical energy to drive a non-spontaneous redox reaction.
Overpotential	Extra voltage above E° needed to discharge a species (esp. gases).
Faraday constant F	Charge of one mole of electrons = 96 500 C mol^{-1}.
Disproportionation	One element simultaneously oxidised and reduced.

A Appendix — Standard Reduction Potentials (25 °C)

All half-equations written as reductions. Strongest oxidant at top (most positive E°); strongest reductant at bottom (most negative E°). Always use the exact values on your official exam data sheet.

Ordered strongest oxidant (top) → strongest reductant (bottom)

Reduction half-equation	E° (V)	Note
$F_2(g) + 2e^- \rightarrow 2F^-$	**+2.87**	strongest oxidant
$H_2O_2 + 2H^+ + 2e^- \rightarrow 2H_2O$	**+1.77**	
$MnO_4^- + 8H^+ + 5e^- \rightarrow Mn^{2+} + 4H_2O$	**+1.51**	acidified permanganate
$Cl_2(g) + 2e^- \rightarrow 2Cl^-$	**+1.36**	
$Cr_2O_7^{2-} + 14H^+ + 6e^- \rightarrow 2Cr^{3+} + 7H_2O$	**+1.33**	acidified dichromate
$O_2(g) + 4H^+ + 4e^- \rightarrow 2H_2O$	**+1.23**	
$Br_2(l) + 2e^- \rightarrow 2Br^-$	**+1.09**	
$Ag^+ + e^- \rightarrow Ag(s)$	**+0.80**	
$Fe^{3+} + e^- \rightarrow Fe^{2+}$	**+0.77**	

$O_2(g) + 2H_2O + 4e^- \rightarrow 4OH^-$	**+0.40**	
$Cu^{2+} + 2e^- \rightarrow Cu(s)$	**+0.34**	
$Sn^{4+} + 2e^- \rightarrow Sn^{2+}$	**+0.15**	
$2H^+ + 2e^- \rightarrow H_2(g)$	**0.00**	**SHE reference**
$Pb^{2+} + 2e^- \rightarrow Pb(s)$	**−0.13**	
$Sn^{2+} + 2e^- \rightarrow Sn(s)$	**−0.14**	
$Ni^{2+} + 2e^- \rightarrow Ni(s)$	**−0.25**	
$Fe^{2+} + 2e^- \rightarrow Fe(s)$	**−0.44**	
$Zn^{2+} + 2e^- \rightarrow Zn(s)$	**−0.76**	used in galvanising
$2H_2O + 2e^- \rightarrow H_2(g) + 2OH^-$	**−0.83**	water reduction
$Al^{3+} + 3e^- \rightarrow Al(s)$	**−1.66**	
$Mg^{2+} + 2e^- \rightarrow Mg(s)$	**−2.37**	
$Na^+ + e^- \rightarrow Na(s)$	**−2.71**	
$Ca^{2+} + 2e^- \rightarrow Ca(s)$	**−2.87**	
$K^+ + e^- \rightarrow K(s)$	**−2.93**	
$Li^+ + e^- \rightarrow Li(s)$	**−3.04**	strongest reductant

CONSTANTS & FORMULAE

$F = 96\,500$ C mol^{-1} · e^- charge $= 1.602\times10^{-19}$ C · $N_A = 6.022\times10^{23}$ mol^{-1} · $V_m = 24.79$ L mol^{-1} (25 °C, 100 kPa).

$Q = It$ · $n(e^-) = Q/F$ · $E°_{cell} = E°_{cathode} - E°_{anode}$ · $\Delta G° = -nFE°_{cell}$.

End of notes — practise balancing half-equations and Faraday calculations under timed conditions.